

Cedar Bluff Quarry Haul Road - Least-Cost Route Report (route CB-HR-01)

Basis: drawing set CB-HR C-001 to C-004 Rev D and the two rasters it registers. Horizontal CRS WGS 84 / UTM zone 14N (EPSG:32614); lengths are horizontal. Status: route study only, not for construction.

1. Result

Quantity	Value
Total route cost, construction + loaded haulage (USD)	3,110,903.74
Construction cost (C-002 rates x grade factor)	2,926,731.02
Loaded haulage cost (C-004 D7: 8,000 USD/m x charged loaded rise)	184,172.72 (23.022 m of rise climbed loaded above 4.0 % effective)
Horizontal length, start cell centre to end cell centre (m)	5,974.53
Vertices (DEM cell centres) / moves / deflection vertices	492 / 491 / 33
Start cell centre (T1 on CR-114)	E 583155.00 N 3349105.00 z 374.54 m
End cell centre (G1 re-surveyed, C-004 Table 6)	E 586455.00 N 3350105.00 z 484.97 m
Creek crossing	Approved window X-2, culvert on 2 watercourse cells at N 3349565
Crossing through the other window (X-1)	None compliant: no route exists through X-1 (Section 5)
Max grade rising / falling in chainage direction (T1 10.0 %)	+9.98 % / -5.88 %
Max loaded uphill grade (D3: 9.0 - 3.0 = 6.0 %)	5.88 %
Longest continuous run of moves > 8.0 % (T3 60.0 m)	56.6 m
Largest change of direction between moves (T4 45 deg)	45 deg
Smallest K between consecutive moves (D5 1.4 m/%)	1.427 m/%
Shortest tangent between deflection vertices (D6: $\geq 2R \tan 22.5 = 37.28$ m)	40.00 m
Start to first / last to end deflection (D6: $\geq R \tan 22.5 = 18.64$ m)	30.00 m / 183.85 m
Escarpment ascent	northern slump; route reaches N 3351475 at E 586025, then returns south along the plateau to G1
Formation edge to wetland W-1 / to HS-1	657.7 m (limit 30.0) / 446.6 m (limit 150.0)

2. Method

Terrain. DEM DN converted with $\text{Elevation(ft)} = 1150.00 + 0.01 \times \text{DN}$ (C-002 Table 1), then to metres with the US survey foot (1200/3937 m). Grades use metres on both axes.

Georeferencing. Both rasters are pixel-is-area with the registered upper-left corner. DEM cell centres are at E 583 005 + 10i, N 3 351 495 - 10j. The land-cover raster has its own origin (E 582 900, N 3 351 620) and 20 m pixels; each DEM cell takes the class of the land-cover pixel containing its centre (exact RGB match).

Control points. T1, X-1, X-2 and HS-1 from C-002 Table 3 (WGS 84) with PROJ. G1 from C-004 Table 6: TxDOT surface coordinates E 3,001,506.17 ft, N 10,073,679.76 ft, divided by the surface adjustment factor 1.00012 to grid E 3,001,146.03 ft, N 10,072,471.06 ft in NAD83 / Texas Central, US survey feet (EPSG:2277), then transformed with PROJ to E 586,454.90, N 3,350,102.50 (cell E 586455, N 3350105). Using the surface values as grid puts G1 about 110 m east and 368 m north; reading the feet as international feet moves it about 6 m south into the next cell; the superseded Rev C position (C-002 Table 3, still plotted on C-001) is two cells west.

Road footprint (C-003). Formation $1.00 + 1.00 + 10.00 + 1.00 + 1.00 = 14.00$ m, 7.00 m either side of the centreline; setback and exclusion measured to the formation edge along every move (centreline clearance > 37.0 m from wetland pixels, > 157.0 m from HS-1).

C-003 Table 4. T1 every move at most 10.0 %; T2/T3 consecutive moves steeper than 8.0 % total at most 60.0 m; T4 direction changes by at most 45 deg between consecutive moves.

C-004 loaded direction (D1-D3). Loaded trucks run G1 to T1, against chainage. A move that falls in chainage direction is climbed loaded; its effective grade is grade + 3.0 % rolling resistance, limited to 9.0 %, so its grade may not exceed 6.0 %. Moves rising in chainage direction are loaded downhill and keep the 10.0 % T1 limit.

Vertical curvature (D5). For consecutive moves of lengths L1, L2 and signed grades g1, g2, the distance between their midpoints along the centreline is $(L1 + L2)/2$, so $|g2 - g1| \leq (L1 + L2) / (2 \times 1.4)$: 7.14 % between two orthogonal moves, 8.62 % orthogonal-diagonal, 10.10 % between two diagonal moves.

Horizontal curves (D6). Every deflection on the grid is 45 deg, so an arc of radius 45 m needs a tangent length $T = R \tan(22.5 \text{ deg}) = 18.64$ m on each side. Arcs may not overlap, so successive deflection vertices must be at least $2T = 37.28$ m apart (4 orthogonal or 3 diagonal moves), and the first and last deflection vertices at least $T = 18.64$ m from the start and end cell centres (2 moves).

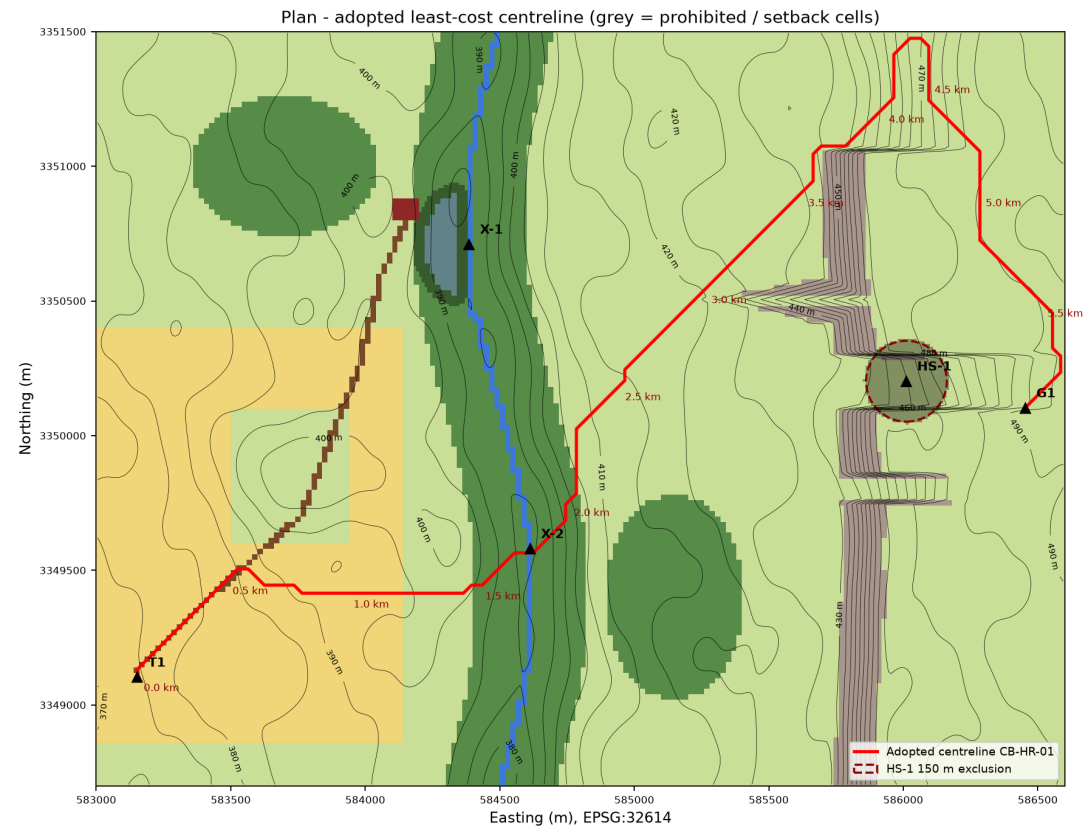
Objective (C-004 Note D1, D7). Move cost = $d \times (c_i + c_j)/2 \times f(|g|) + 8,000 \times (z_i - z_j)$ when the move is climbed loaded (falls in chainage direction) at an effective grade above 4.0 %, i.e. a grade above 1.0 % once the 3.0 % rolling resistance is added; otherwise the construction term alone. The fall is not credited; the 450 USD/m brake-wear and 6.0 % empty top-gear figures in Table 7 are information only.

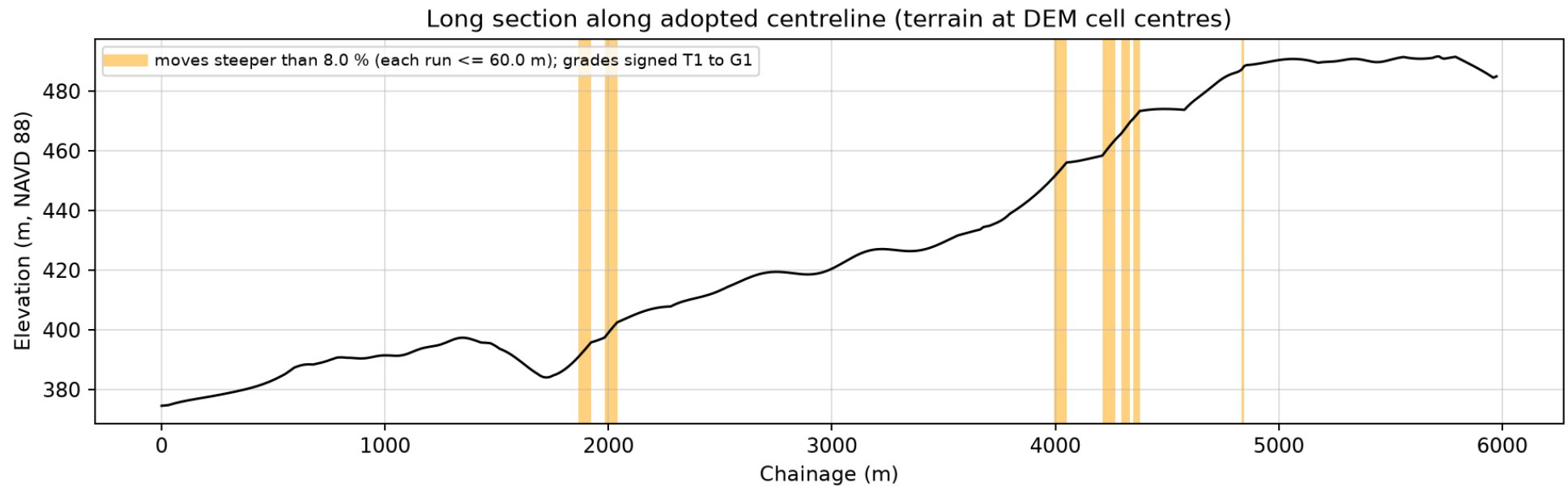
Search. T3, T4, D5 and D6 depend on the path, so the search runs on an expanded state: cell x arrival heading (which also fixes the previous move and its grade for D5) x steep-run length (for T3) x moves since the last deflection vertex, with a separate counter for the first tangent (for D6). Dijkstra on this graph gives the exact optimum; the end state is accepted only if the last tangent is at least 18.64 m.

Rates and constraint values applied (C-002, C-003, C-004 Rev D)

Item applied	Value
Grassland / pasture; cultivated cropland	420 / 480 USD/m
Woodland (Rev B; Rev A 690 withdrawn); rock outcrop	940 / 1,150 USD/m
Existing gravel track; culvert cell within 30.0 m of X-1 / X-2	160 / 3,800 USD/m
Watercourse elsewhere, wetland, farmstead	prohibited
Formation width; wetland setback; HS-1 radius (to formation edge)	14.0 m; 30.0 m; 150.0 m
T1 max grade; T2/T3 steep threshold / run; T4 turn	10.0 %; > 8.0 % / 60.0 m; 45 deg
D1 loaded direction; D2 rolling resistance; D3 effective limit	G1 to T1; 3.0 %; 9.0 % (grade 6.0 %)
D5 minimum K; D6 minimum radius	1.4 m per %; 45.0 m (tangents 37.28 m / 18.64 m)
D7 loaded haulage	8,000 USD per m of loaded rise on moves above 4.0 % effective (1.0 % grade)
Grade factor f(g)	1.00 (g<=3); 1.00+0.05(g-3) (3<g<=6); 1.15+0.12(g-6) (6<g<=10)

3. Plan and long section





4. Breakdown

Land-cover class	Length (m)	Construction cost (USD)
Grassland / pasture	4,215.1	1,865,786
Cultivated cropland	632.3	303,816
Woodland	533.9	587,744
Existing gravel track	573.3	92,731
Watercourse at X-2 (culvert)	20.0	76,654
Total	5,974.5	2,926,731

Each move's length is split equally between its two end cells; its construction cost is split in proportion to the two cells' unit rates. Loaded haulage is not allocated to classes.

Grade band (abs)	0-3 %	3-6 %	6-8 %	8-10 %
Length (m)	3,865.8	1,483.1	317.0	308.7

5. What controls the alignment

Creek crossing - X-1 is not feasible. At X-1 the eastern watercourse column is 35.0 m from the W-1 boundary; the formation edge would be $35.0 - 7.0 = 28.0$ m from the wetland. With X-2 removed the search finds no compliant route, so the only crossing is X-2.

Cedar Bluff escarpment. The saddle gap next to HS-1 is closed by the 150 m exclusion measured to the formation edge; the spur crest grades at about 9.2 % for over 600 m and breaks T3 (60 m); the chute needs 90 and 135 deg switchbacks and breaks T4. The route climbs the northern slump and returns south along the plateau to G1.

C-004. The loaded-uphill limit (6.0 % on moves falling in chainage direction) reshapes the descent into the Cedar Branch valley and the plateau return; applied the wrong way round (to chainage-rising moves) or to both directions it closes every escarpment ascent, so no route exists. The 45 m radius forces tangents of at least 37.28 m between deflections and removes the short zig-zags the unconstrained optimum uses. $K = 1.4$ alters the slump climb where steep runs are broken. Loaded haulage adds 184,173 USD and moves the route to one with less loaded climbing above the top-gear range; because of the 4.0 % threshold, charging the rise in chainage direction, dropping the rolling resistance from the threshold or charging every loaded rise each gives a different line. Each run below breaks exactly one stated rule and is not an option.

Run	Total (USD)	Length (m)	Vertices	Xing	Max loaded uphill %	Min K	Min tangent (m)
Adopted route (all C-002, C-003 and C-004 criteria)	3,110,904	5,974.5	492	X-2	5.88	1.43	40.0
Check: D3 loaded adverse limit ignored (or 9.0 % read as grade)	3,079,905	5,966.2	492	X-2	8.11	1.43	40.0
Check: D5 vertical curvature ignored	3,108,097	5,962.8	490	X-2	5.88	1.35	40.0
Check: D6 curve radius ignored	3,037,459	5,988.1	483	X-2	5.49	1.40	10.0
Check: D6 with R tan(22.5) between deflections	3,050,379	5,997.4	486	X-2	5.49	1.40	20.0
Check: D7 haulage ignored (construction cost only)	2,890,615	6,058.4	495	X-2	5.49	1.44	40.0
Check: D7 rise charged in chainage direction	3,967,261	5,982.8	492	X-2	5.88	1.43	40.0
Check: D7 threshold 4.0 % read as grade (no rolling resistance)	2,964,315	6,094.9	497	X-2	5.49	1.44	40.0
Check: D7 threshold ignored (every loaded rise charged)	3,131,810	5,974.5	492	X-2	5.88	1.43	40.0
Check: G1 at superseded Rev C position	2,890,113	5,623.7	471	X-2	5.88	1.44	40.0
Check: EPSG:2277 read as international feet	3,274,997	6,401.1	533	X-2	5.88	1.43	40.0
Check: G1 surface coordinates used as grid (SAF ignored)	2,862,024	5,552.4	456	X-2	5.88	1.43	40.0
Check: C-004 ignored entirely (Rev C rules)	2,618,852	5,513.4	455	X-2	8.14	1.19	10.0
Check: 45 deg direction-change limit (T4) ignored	2,976,243	5,738.0	470	X-2	5.75	1.45	40.0
Check: 60 m sustained-grade limit (T3) ignored	2,567,344	4,621.3	401	X-2	5.73	1.73	40.0
Check: without HS-1 exclusion	2,217,326	4,000.9	346	X-2	5.73	1.52	40.0
D3 applied to chainage-rising moves / to both directions	no route						
D5 read as a flat 7.14 % grade-change limit	no route						

Only X-1 available	no route						
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6. Compliance and limitations

Every move is at or below 10.0 % (max 9.98 %); loaded uphill at most 5.88 %; the longest run over 8.0 % is 56.6 m; no change of direction exceeds 45 deg; $K \geq 1.427$; tangents ≥ 40.00 m (ends 30.00 / 183.85 m); the formation edge stays 657.7 m from W-1 and 446.6 m from HS-1; the only watercourse cells used are within 30 m of X-2.

The centreline is a 10 m raster least-cost path screened against C-004 geometry; it is a corridor, not a geometric design. Tie-break order, international versus US survey foot for the DEM, and thresholds moved by 0.02 (grade), 0.01 (K) or 1 m (radius) give the same route.

Deliverables: CB_HaulRoad_Centreline.gpkg (layer 'centreline', one LineString, EPSG:32614), CB_HaulRoad_Vertices.csv (one row per vertex, start to end; cum_cost_usd is the total route cost including haulage), this report.